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Partitions and their generating functions

DEFN An (integer) partition $\lambda = (\lambda_1, \lambda_2, \lambda_3, \dots)$ of n is a • weakly decreasing, eventually zero, $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \dots \geq 0$ sequence of nonnegative integers with $\lambda_1 + \lambda_2 + \lambda_3 + \dots = n$.

We write $\lambda \vdash n$ (' λ divides n ') and $|\lambda| = n$. 'size' of λ

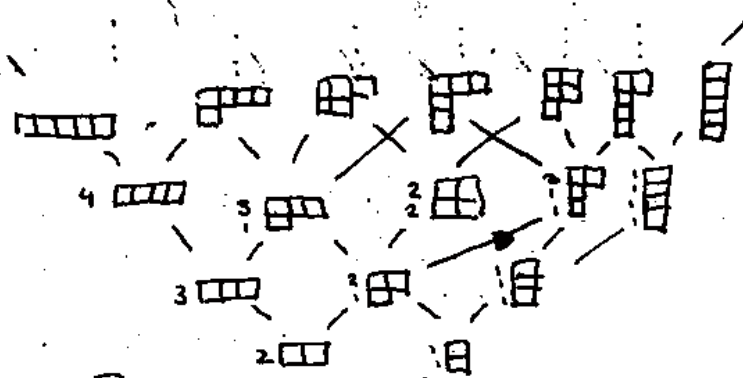
e.g. $\lambda = (5, 5, 3, 1, 0, 0, \dots) = (5, 5, 3, 1, 0) = (5, 5, 3, 1) \vdash 14 = 5+5+3+1$

Its length $\ell(\lambda) := \#\{i: \lambda_i > 0\} = \#$ of nonzero parts λ_i .

Its Young diagram is a left + top justified array of boxes, with λ_i boxes in the i th row from the top.

e.g. $\lambda = (5, 5, 3, 1) \leftrightarrow$ 

Let $p(n) := \#$ of partitions $\lambda \vdash n$



$p(n)$	n
7	5
5	4
3	3
2	2
1	1
1	0

$\emptyset$ (empty partition, unique partition of 0)

$\mathbb{Y} =$ Young's lattice, the poset of all partitions, ordered by containment of Young diagrams
 $\lambda \nearrow \mu$ means λ obtained from μ by adding one box

$$\sum_{n \geq 0} p(n) q^n = \sum_{\lambda \text{ all partitions}} q^{|\lambda|} = \left[\begin{array}{cc} 1 + q + q^2 + q^3 + \dots & 1 + q^2 + q^4 + q^6 + \dots \\ 1 + x_1 + x_1^2 + x_1^3 + \dots & 1 + x_2 + x_2^2 + x_2^3 + \dots \\ \vdots & \vdots \end{array} \right]$$

$$= \left[\begin{array}{c} \begin{array}{c} x_1^4 \\ x_1^3 x_2 \\ x_1^2 x_2^2 \\ x_1 x_2^3 \\ x_2^4 \end{array} \\ \vdots \end{array} \right] = x_1^2 x_2^3 x_3^0 x_4^1 = q^2 q^6 q^0 q^4 = q^{12}$$

$$\begin{aligned} x_i &\mapsto q^i \\ &= (1 + q + q^2 + \dots) (1 + q^2 + q^4 + \dots) (1 + q^3 + q^6 + \dots) \\ &= \frac{1}{1-q} \cdot \frac{1}{1-q^2} \cdot \frac{1}{1-q^3} \cdot \dots \text{ a convergent product} \\ &= \prod_{n=1}^{\infty} \frac{1}{1-q^n} \leftarrow \text{g.f. for all partitions, as a product formula!} \end{aligned}$$

Cultural asides:

Thm (Euler's pentagonal number theorem) "pentagonal numbers"

$$\prod_{n=1}^{\infty} (1 - q^n) = 1 + \sum_{k=1}^{\infty} (-1)^k (q^{k(3k+1)/2} + q^{k(3k-1)/2})$$

$$= 1 - q - q^2 + q^5 + q^7 - q^{12} - q^{15} + q^{22} + q^{26} - \dots$$

Pf: We will give a bijective proof later in the course. $\square$

Since $\sum p(n) q^n = (\prod_{n=1}^{\infty} (1 - q^n))^{-1}$ Pent. # Thm implies a recurrence for $p(n) = \sum_{k=-\infty, k \neq 0}^{\infty} (-1)^k p(n - k(3k-1)/2)$

$$= p(n-1) + p(n-2) - p(n-5) - p(n-7) + p(n-12) + p(n-15) - \dots$$

Thm (Hardy + Ramanujan) $p(n) \sim \frac{1}{4n\sqrt{3}} e^{\pi\sqrt{\frac{2n}{3}}}$

Pf: "Circle method," careful analysis of the singularities of $\prod \frac{1}{1-q^n}$ thought of as an analytic function. Like what we saw with rational g.f. of Fibonacci #s, but much more hardcore complex analysis. $\square$

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G.f.'s for restricted classes of partitions:

Let $d(n) := \#$ of partitions of n into distinct parts.

n	$d(n)$	
0	1	$\emptyset$
1	1	$\square$
2	1	$\square \quad \square$
3	2	$\square \quad \square \quad \square$
4	2	$\square \quad \square \quad \square \quad \square$
5	3	$\square \quad \square \quad \square \quad \square \quad \square$

$$D(q) := \sum_{n \geq 0} d(n) q^n = (1+q)(1+q^2)(1+q^3)(1+q^4) \cdots = \prod_{j=1}^{\infty} (1+q^j)$$

Let $o(n) := \#$ partitions of n into odd parts.

n	$o(n)$	
0	1	$\emptyset$
1	1	$\square$
2	1	$\square$
3	2	$\square \quad \square$
4	2	$\square \quad \square$
5	3	$\square \quad \square \quad \square$

Looks like possibly $o(n) = d(n)$, but how to show it?

G.f.'s!

$$\begin{aligned} O(q) &:= \sum_{n \geq 0} o(n) q^n = (1+q+q^2+\cdots)(1+q^3+q^6+\cdots) \\ &\quad (1+q^5+q^{10}+\cdots) \\ &= \frac{1}{1-q} \cdot \frac{1}{1-q^3} \cdot \frac{1}{1-q^5} \cdots = \prod_{j=1}^{\infty} \frac{1}{1-q^{2j+1}} \\ &\stackrel{?}{=} \prod_{j=1}^{\infty} (1+q^j) \end{aligned}$$

How to show $D(q) = O(q)$?

Well, $D(q) = (1+q)(1+q^2)(1+q^3) \dots$

$$= \frac{(1-q^2)}{(1-q)} \frac{(1-q^4)}{(1-q^2)} \frac{(1-q^6)}{(1-q^3)} \dots$$

Recall: $\frac{(1+x)(1-x)}{(1-x^2)}$

$$= \frac{(1-q^2)(1-q^4)(1-q^6)(1-q^8) \dots}{(1-q)(1-q^2)(1-q^3)(1-q^4) \dots}$$

$$= \frac{1}{(1-q)(1-q^3)(1-q^5) \dots} = O(q) !$$

Was that manipulation ok? Yes! Thinking slightly differently...

Let $R(q) := (1-q)(1-q^3)(1-q^5) \dots = \frac{1}{O(q)} \in \mathbb{C}[[q]]$

Want to show $1 \stackrel{?}{=} D(q) R(q)$ in $\mathbb{C}[[q]]$

$$1 + 0 \cdot q + 0 \cdot q^2 + \dots$$

$$= ((1+q)(1+q^2)(1+q^3) \dots) ((1-q)(1-q^3)(1-q^5) \dots)$$

$$= \underbrace{(1+q)(1-q)}_{(1-q^2)} \underbrace{(1+q^2)(1-q^3) \dots (1+q^3)(1-q^5) \dots}_{(1-q^4)(1-q^6) \dots}$$

$$= (1-q^4) \underbrace{(1+q^3)(1+q^4) \dots (1+q^3)(1-q^5) \dots}_{(1-q^5)(1-q^7) \dots}$$

$$= (1-q^4)(1-q^6) \underbrace{(1+q^4)(1+q^5) \dots (1+q^5)(1-q^7) \dots}_{(1-q^7)(1-q^9) \dots}$$

$$= 1 + 0 \cdot q + 0 \cdot q^2 + 0 \cdot q^3 + \dots$$

$$= \dots \text{ et cetera}$$

hinted at by this manipulation

Note: $\exists$ bijective proof that $d(n) = o(n)$ as well

(See Stanley Prop. 1.8.5) for 2 such proofs... Basic idea is binary expansion:

e.g. $\lambda = (9^5, 5^{12}, 3^2, 1^3) = (9^{2^0+2^2}, 5^{2^2+2^3}, 3^{2^1}, 1^{2^0+2^1})$

$O(n)$ $(9, 9, 9, 9, 9, 5, 5, \dots, 5, 3, 3, 1, 1, 1)$

$n=114$

$\leftrightarrow \mu = (9 \cdot 2^0, 9 \cdot 2^1, 5 \cdot 2^2, 5 \cdot 2^3, 3 \cdot 2^1, 1 \cdot 2^0, 1 \cdot 2^1)$

$= (9, 36, 20, 40, 6, 1, 2)$

(reorder)

$= (40, 36, 20, 9, 6, 2, 1) \in d(n)$

//

Compositions and their generating functions

DEF'N A composition $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_k)$ of n , denoted $\alpha \models n$, is a sequence of positive integers $\alpha_i \in \mathbb{Z}_{1,2,3,\dots}$ w/ $\alpha_1 + \alpha_2 + \dots + \alpha_k = n$. (Unlike a partition, the α_i need not be weakly decreasing.) As w/ partitions, we call α_i the parts of the composition.

e.g. $\alpha = (1, 4, 2, 4)$ is a composition of 11 into 4 parts.

Let $\bar{c}_k(n)$ denote # compositions of n into k parts and $\bar{c}(n)$ denote # compositions of n (into any # of parts).

Prop. $\sum_{n=0}^{\infty} \bar{c}_k(n) \cdot x^n = \left(\frac{1}{1-x} - 1\right)^k = \left(\frac{x}{1-x}\right)^k$

Pf: Note $\left(\frac{1}{1-x} - 1\right) = 1 + x + x^2 + x^3 + \dots - 1 = x + x^2 + x^3 + \dots$

Now use "picture-writing":

e.g. $k=3$

Cor. $\sum_{n=0}^{\infty} \bar{c}(n) \cdot x^n = 1 + \frac{x}{1-2x}$

Pf: Note that $\bar{c}(n) = \sum_{k=0}^{\infty} \bar{c}_k(n)$, so

$$\begin{aligned} \sum_{n=0}^{\infty} \bar{c}(n) \cdot x^n &= \sum_{n=0}^{\infty} \left(\sum_{k=0}^{\infty} \bar{c}_k(n) \right) \cdot x^n = \sum_{k=0}^{\infty} \left(\sum_{n=0}^{\infty} \bar{c}_k(n) \cdot x^n \right) \\ &= \sum_{k=0}^{\infty} \left(\frac{x}{1-x} \right)^k = \frac{1}{1 - \frac{x}{1-x}} \\ &= \frac{1-x}{1-2x} = 1 + \frac{x}{1-2x} \end{aligned}$$

Cor. For $n \geq 1$, $\bar{c}(n) = 2^{n-1}$.

Pf. $1 + \frac{x}{1-2x} = 1 + \sum_{n \geq 0} 2^n x^{n+1} = 1 + \sum_{n \geq 1} 2^{n-1} x^n$ $\square$
Then extract coeff. of x^n in previous cor.

Since this is such a simple formula, we could ask for a direct proof, not using generating functions. In fact...

Prop. The # of compositions of n into k parts is $\bar{c}_k(n) = \binom{n-1}{k-1}$.

Pf. Let's say a sequence $\alpha = (\alpha_1, \dots, \alpha_k)$ of nonnegative integers $\alpha_i \in \{0, 1, \dots\}$ w/ $\sum \alpha_i = n$ is a weak composition of n .

Claim # of weak compositions of n into k parts $= \left(\binom{k}{n} \right) = \binom{k+n-1}{n}$ $\leftarrow$ recall 'multichoose' #

Pf. Write a weak composition of n , e.g.
 $\alpha = (2, 0, 1, 3)$ using 'stars and bars'
as $\alpha = **|1*|*****$.

We saw before that these patterns are counted by $\left(\binom{k}{n} \right)$.

Finally, $\exists$ a bijection $\{\text{weak comp. of } n \text{ into } k \text{ parts}\} \leftrightarrow \{\text{(usual) comp. of } n+k \text{ into } k \text{ parts}\}$
 $(\alpha_1, \alpha_2, \dots, \alpha_k) \mapsto (\alpha_1+1, \alpha_2+1, \dots, \alpha_k+1)$.

Hence # comp. of n into k parts $= \left(\binom{k}{n-k} \right) = \binom{k+n-k-1}{n-k} = \binom{n-1}{n-k} = \binom{n-1}{k-1}$ $\square$

Cor. # comp. of n is $\bar{c}(n) = 2^{n-1}$ for any $n \geq 1$.

Pf. $\bar{c}(n) = \sum_{k=1}^n \binom{n-1}{k-1} = \sum_{k=0}^{n-1} \binom{n-1}{k} = 2^{n-1}$ $\checkmark$ $\square$

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Some useful formal power series

Let's define and study some specific elements of $\mathbb{C}[[x]]$:

DEFN $e^x := \sum_{n \geq 0} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$

$\log(1+x) := \sum_{n \geq 1} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$

$\forall \lambda \in \mathbb{C}, (1+x)^\lambda := \sum_{k \geq 0} \binom{\lambda}{k} x^k$

where $\binom{\lambda}{k} := \frac{\lambda(\lambda-1)(\lambda-2)\dots(\lambda-(k-1))}{k!} \in \mathbb{C}$ "generalized binomial coefficient"

(just like for $n \in \mathbb{N}$, $\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1)\dots(n-(k-1))}{k!}$)

These formal power series satisfy the properties you'd expect:

E.g. ① $(1+x)^\lambda (1+x)^\mu = (1+x)^{\lambda+\mu} \in \mathbb{C}[[x]] \quad \forall \lambda, \mu \in \mathbb{C}$

② $e^x e^y = e^{x+y} \in \mathbb{C}[[x, y]]$

③ $e^{\log(1+x)} = 1+x, \quad e + \dots$

defined to be $= 1 + \log(1+x) + \frac{\log(1+x)^2}{2!} + \dots$
 $= 1 + (x - \frac{x^2}{2} + \frac{x^3}{3} - \dots) + \frac{(x - \frac{x^2}{2} + \frac{x^3}{3} - \dots)^2}{2} + \dots$

Why does this even converge in $\mathbb{C}[[x]]$?

Prop. If $A(x) = \sum_{n \geq 0} a_n x^n$, $B(x) = \sum_{n \geq 0} b_n x^n$, and $b_0 = 0$, then $A(B(x)) := \sum_{n \geq 0} a_n (B(x))^n$ converges in $\mathbb{C}[[x]]$.

How to justify ①, ②, ③, etc...? Could do a tedious manipulation of coefficients, but instead, since e^x , $\log(1+x)$, $(1+x)^\lambda$ are also analytic functions we are familiar with (whose power series expansions are as above), we can use a trick from complex analysis...

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④ for $n \in \mathbb{N}$, $(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k = \sum_{k=0}^n \binom{n}{k} x^k$,

but also $\frac{1}{(1-x)^n} = (1+(-x))^{-n} = \sum_{k=0}^{\infty} \frac{(-n)(-n-1)(-n-2)\dots(-n-(k-1))}{k!} (-x)^k$

$(1+x+x^2+\dots)(1+x+x^2+\dots)\dots(1+x+x^2+\dots)$

 $\swarrow \quad \quad \quad \searrow$

 $**|*||\dots|*|**$ e.g. $n=18$

 $\nearrow$ flavors of bagels

PS that $\left(\binom{n}{k}\right) = \binom{n+k-1}{k} \cdot \left(\text{'n multichoose k'}\right) \left(\binom{n}{k}\right) := \# \text{ k element multisets w/ entries in } \{1, 2, \dots, n\}$

Stars and bars'

Stars indicate how many of each element is chosen, bars separate bins for each element

e.g. $\{1, 1, 2, 4, 4, 4, 7\}$ is
a 7-element multisubset of $\{1, 2, \dots, 9\}$

$$\text{9/5} \quad \textcircled{5} \quad \frac{1}{1-4x} = \sum_{k \geq 0} \binom{-1}{k} (-4x)^k = \sum_{k \geq 0} \underbrace{\binom{1+k-1}{k}}_{=(k)-1} 4^k x^k = \sum_{k \geq 0} 4^k x^k \checkmark$$

but also $\frac{1}{(1-4x)^2} = \sum_{k=0}^{\infty} \underbrace{\binom{2+k-1}{k}}_{\binom{k+1}{k} = k+1} 4^k x^k = \sum_{k=0}^{\infty} (k+1) 4^k x^k$

$$\frac{1}{(1-4x)^3} = \sum_{k=0}^{\infty} 4^k x^k \cdot \binom{k+2}{2}, \quad \frac{1}{(1-4x)^4} = \dots \text{ etc.}$$

Useful for extracting coefficients of a rational function after performing the Partial fraction expansion . . .

$$\begin{aligned}
 \textcircled{6} \quad \frac{1}{\sqrt{1-4x}} &= (1-4x)^{-1/2} = \sum_{k \geq 0} \binom{-1/2}{k} (-4)^k x^k \\
 &= \sum_{k \geq 0} \left(\frac{(-1/2)(-3/2)(-5/2) \cdots (-(2k-1)/2)}{k!} (-4)^k \right) x^k \\
 &= \sum_{k \geq 0} \left(\frac{4^k (1)(3)(5) \cdots (2k-1)}{2^k k!} \right) x^k \\
 &= \sum_{k \geq 0} \left(\frac{2^k (1)(3)(5) \cdots (2k-1)}{k!} \frac{(2)(4)(6) \cdots (2k)}{2^k k!} \right) x^k \\
 &= \sum_{k \geq 0} \frac{(2k)!}{k! \cdot k!} x^k = \sum_{k \geq 0} \binom{2k}{k} x^k
 \end{aligned}$$

this F.P.S.
 is
algebraic
 but not
 rational

central
 binomial
 coefficients...

interesting...

Another tool from calculus that's useful for $R[[X]]$

DEFIN For $A(x) = \sum_{n \geq 0} a_n x^n \in R[[X]]$, we define the formal derivative $A'(x) := \sum_{n \geq 1} \underbrace{n \cdot a_n}_{\substack{\text{"} \\ a_n + a_n + \dots + a_n \\ \text{"} \\ n \text{ times}}} x^{n-1} \in R[[X]]$

The derivative satisfies the usual rules from calculus:

- $(A(x) + B(x))' = A'(x) + B'(x)$
- $(AB)' = A' \cdot B + B' \cdot A$
- $\left(\frac{1}{A}\right)' = \frac{-A'}{A^2}$
- $(A(B(x)))' = A'(B(x)) \cdot B'(x)$
- etc...